

What AI Presence Does Not Predict

Amazon Best Sellers Rank as discriminant validity evidence for the AIAS construct

May 2026 · Pablo Ulpiano González Castro

AIAS™ Measurement Program · Discriminant Validity

AI Presence scores show zero correlation with Amazon Best Sellers Rank across three consumer-goods categories. Brands that language models recognize are not the same brands that dominate retail sales. This is not a failure of the measure — it is evidence that AI Availability captures something genuinely new.

$\rho \approx 0.000$ Pooled correlation · **0 / 3**
Substrates with signal · **104 / 106** Amazon
coverage · **3 orders** BSR range

systems. These are independent dimensions of brand access — exactly what the Tri-System framework predicts.

This study tested whether brands that score high on AI Presence also sell well on Amazon. They do not. Across kitchen knives, audiophile headphones, and skincare, the correlation between AI Presence Index (C_p) and Amazon Best Sellers Rank was indistinguishable from zero — not merely non-significant, but effectively absent.

The null result is the finding. Combined with the strong positive correlation between C_p and Google Trends search interest found in v0.25 ($\rho = 0.74$), this study completes a convergent–discriminant validity pair: AI Presence tracks brand salience in AI knowledge systems (converges with search interest) but does not track commercial outcomes (diverges from retail sales rank).

For brand strategists, the implication is clear: AI Availability is not a proxy for market share. A brand can be universally recognized by language models and still rank poorly on Amazon, or dominate Amazon sales while being invisible to AI

What we measured

We correlated two signals across 106 brands in five consumer-goods categories.

AI Presence Index (C_p): How many of six major language models recognize each brand in its category (integer 0–6). Three substrates provided within-category variance: kitchen knives (26 brands, v0.16 retrofitted), audiophile headphones (16 brands, v0.19), and skincare (24 brands, v0.20). A fourth substrate — cosmetics (v0.21) — showed a ceiling effect where all 24 brands scored $C_p = 6$, providing a natural control. A fifth substrate (premium kitchenware, v0.17) was excluded because its Phase A data was incomplete.

Amazon Best Sellers Rank (BSR): The sales-velocity rank of each brand's best-selling product on Amazon, acquired via automated browser extraction in a single 48-hour window. Lower rank = more sales. Coverage: 104 of 106 brands listed (98.1%).

Test: Spearman rank correlation per substrate (does higher C_p predict lower/better BSR?), plus pooled cross-substrate correlation and a Cell A separation test.

FINDING 01

Zero correlation, uniformly

AI Presence does not predict Amazon BSR

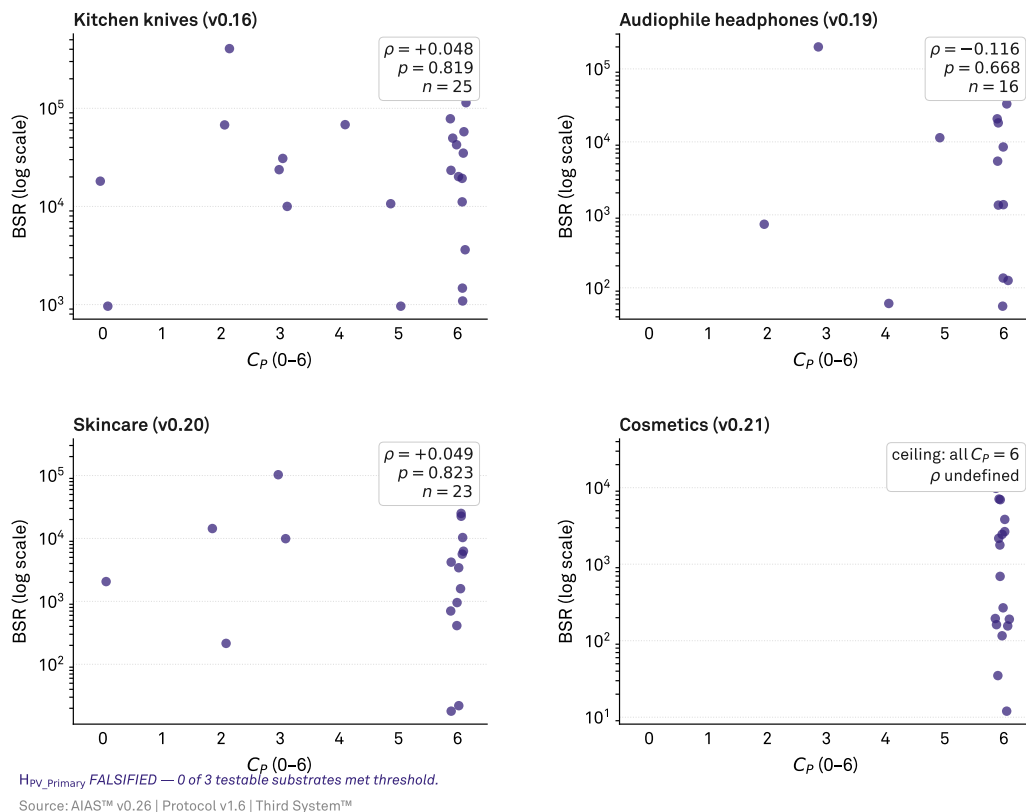
 C_P (0–6) vs. Amazon Best Sellers Rank by substrateListed brands only; one dot per brand. Spearman ρ tests within-substrate monotonicity.

Figure 1 · AI Presence does not predict Amazon BSR. C_P (0–6) vs. Amazon Best Sellers Rank by substrate. Each panel shows listed brands with Spearman ρ annotated. Kitchen knives, audiophile headphones, and skincare show indistinguishable-from-zero correlations; cosmetics shows the ceiling effect (all $C_P = 6$).

Within-substrate Spearman correlations: kitchen knives $\rho = +0.048$ ($p = 0.819$), audiophile headphones $\rho = -0.116$ ($p = 0.668$), skincare $\rho = +0.049$ ($p = 0.823$). Pooled across

88 brands: $\rho = -0.0002$ ($p = 0.998$). The correlations are not merely non-significant — they are effectively zero.

FINDING 02

The ceiling proves the point

In cosmetics, all 24 brands achieved $C_p = 6$ (full AI recognition), yet BSR ranged from 12 (Maybelline) to 29,364 (Anastasia Beverly Hills). Full AI recognition is compatible with

any level of retail performance. The two constructs are independent.

FINDING 03

Convergent + discriminant = construct validity

v0.25 showed C_p correlates strongly with Google Trends search interest ($\rho = 0.74$). This study shows C_p does not correlate with Amazon retail performance ($\rho \approx 0$). Together,

these satisfy the Campbell–Fiske requirements: AI Presence converges with related salience measures and diverges from unrelated commercial outcomes.

FINDING 04

Different data-generating processes

LLM training corpora overrepresent editorial and enthusiast discourse; Amazon BSR reflects pricing, distribution logistics, Prime eligibility, and advertising spend. A brand can dominate

knife forums (high C_p) without dominating Amazon sales, and vice versa. The signals originate in fundamentally different systems.

Limitations

BSR captures sales velocity within one retail channel (Amazon), not total market performance. Brands with strong offline or DTC distribution may appear weaker in BSR than their true market position warrants. A multi-channel sales metric would provide a stronger discriminant validity test.

The v0.16 kitchen knives C_p was retrofitted under the v1.4+ protocol because the original v1.2 methodology did not produce comparable scores. The retrofit aligns the metric but introduces a provenance asymmetry documented in DEVIATIONS Entry 0.

Three testable substrates is the minimum for cross-substrate generalization. The v0.17 exclusion and v0.21 ceiling effect were not anticipated at pre-registration.

What's next

Extend discriminant validity testing to additional commercial outcome measures: Sephora rankings, B&H; Photo bestsellers, and specialty retailer data. Prioritize behavioral correlates (click-through rates, recommendation acceptance) as convergent measures that sit closer to the theorized mechanism of AI Availability. Expand the Campbell–Fiske matrix with additional convergent measures (Wikipedia pageviews, news-corpus brand mentions) and discriminant measures (brand equity surveys, stock price).

Hypothesis scoring

Four pre-registered propositions tested across the v0.26 cross-substrate panel (kitchen knives, audiophile headphones, skincare, cosmetics).

H	Pre-registered prediction	Result	Status
P1	Per-substrate predictive validity	0 of 3 substrates met the $\rho \geq -0.40$ threshold.	FALSIFIED
P2	Cross-substrate pooled correlation	Pooled $\rho = -0.0002$, $p = 0.998$. Indistinguishable from zero.	FALSIFIED
P3	Cell A separation	Median BSR percentile: Cell A = 51.09, Other = 54.00. Negligible and non-significant.	FALSIFIED
P4	Absence–Presence alignment	Only 2 brands Amazon-absent (Nogent, Clé de Peau Beauté). Below the pre-registered minimum of 5 for inferential testing.	UNDETERMINED

Detailed hypothesis results

P1 — Per-substrate predictive validity: FALSIFIED Kitchen knives: $\rho = +0.048$, $p = 0.819$ ($n = 25$) Audiophile headphones: $\rho = -0.116$, $p = 0.668$ ($n = 16$) Skincare: $\rho = +0.049$, $p = 0.823$ ($n = 23$) Cosmetics: ceiling effect (all $C_p = 6$), ρ undefined

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P2 — Pooled correlation: FALSIFIED 88 listed brands, percentile-normalized BSR: $\rho = -0.0002$, $p = 0.998$

P3 — Cell A separation: FALSIFIED $C_p \geq 4$ ($n = 78$): median BSR percentile = 51.09 $C_p < 4$ ($n = 10$): median BSR percentile = 54.00 Mann-Whitney U: not significant

P4 — Absence–Presence alignment: UNDETERMINED Only 2 absent brands across all substrates. Insufficient sample for the pre-registered Mann-Whitney U test (minimum $n = 5$).

About the Third System

The Third System describes a shift in how consumers make decisions: instead of browsing and comparing options directly, many now rely on AI systems to generate recommendations. In this environment, brands compete not only for human attention but for inclusion within AI-generated outputs. This creates a new competitive layer where visibility is determined by how AI systems interpret and surface brands.

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Third System™ (research entity)

Methodology

This report presents findings from Third System's AI Availability research program. The AI Availability Score (AIAS) is an early-stage measurement framework introduced in Tri-System Brand Growth (Gonzalez Castro, 2026). Findings should be used directionally. Construct validity — the correlation between AI Availability and consumer behavior — has not yet been established and is the subject of Phase 3 of the research program. Methodology is published in full and versioned at thirdsystem.ai/methodology (current: v1.6). Registry construction is curated and reflects the framework's view of each category's competitive set; revisions are documented in the methodology log.

Citation

González Castro, P. U. (2026). *What AI Presence Does Not Predict: Amazon Best Sellers Rank as Discriminant Validity Evidence for the AIAS Construct (v0.26)*. Third System. thirdsystem.ai/v26 (SSRN TBD)

Underlying datasets

Phase A recognition: osf.io/ec6wh/v26/data/v26_phase_a.csv

Phase B recall: osf.io/ec6wh/v26/data/v26_phase_b.csv

MLC coding: osf.io/ec6wh/v26/data/v26_mlc_coding.csv

Methodology log: v1.6 (SSRN 6816340).

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